

Prakhar Shekhar Parthasarthi

Full-Stack Software Engineer · Java & Spring Boot · React · AWS

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47% P90 LATENCY CUT	+33% CONVERSION LIFT	195k GET RPS · JAVA REDIS	20 → 5 ON-CALL / SHIFT
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SUMMARY

Full-stack software engineer with **3+ years at Amazon** across Payments and Travel — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture`, bounded `ExecutorService`, `ConcurrentHashMap`), idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure in **AWS CDK (Java)** — paired with **React (SSR & CSR)** + **Redux** for customer-facing flows.

CORE TECHNICAL STACK

JAVA & JVM	Java 17, Spring Boot, Spring MVC, Guice, JUnit 5, Mockito, SLF4J, Maven, Gradle · <code>ExecutorService</code> , <code>CompletableFuture</code> , <code>ConcurrentHashMap</code> , JVM profiling, GC analysis
DISTRIBUTED SYSTEMS	Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry/backoff, DLQ, at-least-once, TDD
AWS	Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java)
FRONT-END	React.js (SSR & CSR), Redux, HTML/CSS, JavaScript/TypeScript
DATA & STORAGE	DynamoDB, Redis, NoSQL modelling, SQL, caching strategies
TOOLING & PROCESS	Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile/Scrum, on-call, code review, incident response · Python, C++

EXPERIENCE

Software Development Engineer Jul 2022 – Sep 2025

Amazon · Travel (Hotels & Flights) · Bengaluru

- Service ownership — Spring Boot, Spring MVC. Owned **4 customer-facing microservices** powering the hotel booking platform — search, listing, room selection, order review — across design, code review, deploy, and on-call. REST controllers, service-layer logic, `@Repository` DynamoDB DAOs via Spring DI. Prime-exclusive discount and partner-discount integration — **~30 bps** funnel churn drop.
- JVM concurrency — `CompletableFuture`, bounded executors. Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 cut **47%** (**1.1s → 0.59s**), conversion up **+33%**. GZIP on 150–200 MB catalog payloads cut wire size **95%**.
- Event-driven — SQS + Step Functions. Designed idempotent Java webhook for real-time flight schedule changes — SQS ingestion, DynamoDB dedup keys, audit logging, automated offline modifications — **43 bps CPT**. Orchestrated booking state machine (confirm/cancel/retry) on Step Functions with Java Lambda handlers.
- Infrastructure as code — AWS CDK (Java). Authored all service IaC in **Java CDK** — Lambdas, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch alarms. Migrated 2 services from hand-rolled CloudFormation — rollback incidents halved.

Jul 2022 — Sep 2025 • Travel (Hotels & Flights) • Bengaluru

- **Testing** — **JUnit 5**, **Mockito**, **Spring Boot**. Held **80%+** line coverage; Mockito argument captors for downstream contract verification; `@SpringBootTest` integration tests against in-memory DynamoDB. Introduced Gradle JaCoCo coverage gates in CI.
- **Operational excellence**. On-call tickets cut from **20+** to **~5 per shift** (**~2 SDE-weeks/week** reclaimed). Java email-automation deprecated stale WebLab experiments — config-hygiene incidents from **5–6** to **0**. Resolved **150+** production incidents; authored COEs. Built auto-insurance purchase flow (Java + React) — **60%+** renewal lift.

Software Development Engineer, Intern

May 2021 – Jul 2021 • Feb 2022 – Jul 2022

Amazon • Bengaluru

- Migrated legacy GraphQL resolvers to **Java** — DynamoDB via Spring-injected DAOs, API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance using past-purchase data — async Java pre-fill workflow.
- Applied OO design and **Google Guice DI**; **90%+** unit-test coverage (JUnit 5 + Mockito).

SELECTED PROJECTS

AI Visual Code Review github.com/praxstack/ai-visual-code-review • npm <code>Node.js</code> <code>TypeScript</code> <code>Jest</code> <code>VS Code API</code> Security-hardened visual code-review tool with GitHub-style diff UI, AI markdown exports, command-injection protection, rate limiting. Published npm + VS Code extension. Published: npm v2.4.1 • MIT.	Redis Server in Java github.com/praxstack/redis-server-java <code>Java 17</code> <code>ConcurrentHashMap</code> <code>JUnit 5</code> RESP2-compatible Redis from scratch; bounded <code>ExecutorService</code>, lock-free reads, hybrid TTL eviction, TCP integration tests. GET 195k rps • SET 154k rps • p50 < 0.3 ms • 42 tests.
Audio Transcription Pipeline github.com/praxstack/audio-transcription-pipeline <code>Python</code> <code>Bedrock</code> <code>pyannote</code> <code>pytest</code> Multi-stage pipeline: raw audio/video → speaker-labelled Markdown. pyannote diarization + AWS Bedrock (Voxtral ASR + Claude repair). 63 tests • 6 stages • MIT.	Coach Atlas github.com/praxstack/coach-atlas <code>React</code> <code>TypeScript</code> <code>Mermaid</code> <code>KaTeX</code> AI technical-interview mentor + tutorial studio; Socratic prep, Mermaid/KaTeX/code rendering, multi-provider BYOK (keys in browser). React + TypeScript • deployed on Vercel.

Open-Source Contributions

Merged into danielmiessler/Fabric (Go, 42k★):

- #2044 — AWS Bedrock bearer-token (ABSK) auth + guided `--setup`. +882 / -83
- #2052 — dynamic Bedrock region fetching (40+ regions). +252 / -15
- #2061 — streaming-goroutine deadlock fix in `Chatter.Send`. +494 / -42

Under review: deer-flow, claude-mem, tolaria.

EDUCATION

M.C.A., Computer Science • MNNIT, Prayagraj	GPA 9.25	2019 – 2022
B.Sc. (Hons), Computer Science • Bhaskaracharya College, University of Delhi		2015 – 2018

AWARDS & RECOGNITION

Amazon Pay Merchants Categories Champion — 100+ hotel trust-buster bugs; on-call 20+ to ~5 per shift.	2023
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